PROTOCODUS

YouKnow

Circuit-Modelled Synth for Reason 14+
User Guide - 1.0.0f2

Direct subtractive synthesis, 77 original patches, polyphonic MIDI, monophonic Note/Gate CV, stereo chorus, and scalable rear-panel engine quality.

Install and update

Install and update YouKnow through your Reason Studios account and the Reason Rack Extension browser. The Universal 45 development archive is for Reason Studios' build service and is not a customer installer.

After installation, find **YouKnow Circuit-Modelled Synth** under Instruments in the device palette. Creating it makes a sequencer track and auto-routes its stereo output when a suitable destination is available.

First sound

1. Create YouKnow and play MIDI notes. The Init patch is loaded by default.
2. Use the patch display and browse controls to choose from 77 supplied original patches, including Init. Sounds are organized into Bass, Leads, Keys, Brass, Pads, Strings, and Effects.
3. Follow the front panel from LFO and DCO through HPF, VCF, VCA, ENV, and CHORUS. Performance and keyboard controls occupy the lower row; Unit Character and Aging are on the rear.
4. Save edited sounds with Reason's normal patch command. Unit Character is stored with the patch; Engine Quality choices and Aging are stored with the song.

Front panel

LFO sets modulation rate and delayed onset. **DCO** selects octave range, waveforms, PWM source and depth, sub level, noise, and oscillator modulation. **HPF** provides four high-pass positions including bass boost.

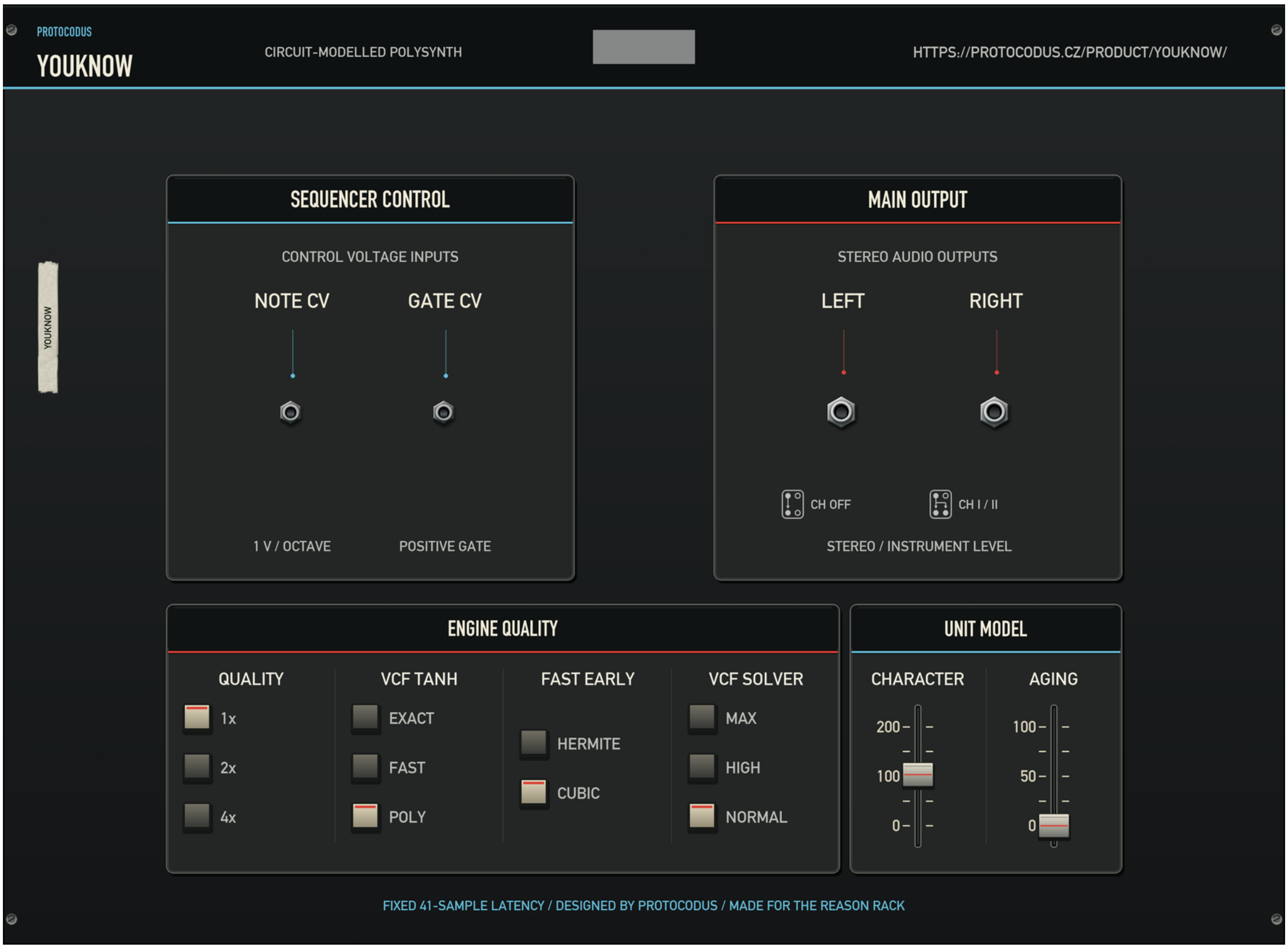
VCF controls cutoff, resonance, envelope polarity and amount, LFO amount, and keyboard tracking. At high resonance the filter can self-oscillate. **VCA** chooses envelope or gate behavior and sets output level. **ENV** is the shared attack, decay, sustain, and release envelope. **CHORUS** offers Off, I, II, and a dedicated narrow I+II combination.

The lower row's Performance bay contains the pitch and modulation wheels, volume, and performance modulation depths. Its Keyboard bay contains glide, key mode, transpose, master tune, velocity response, and 1-16 voices. Chorus Hiss sits in the Chorus bay beside its four modes. Init and patches that inherit the engine default use the quieter MN3009 typical-S/N estimate; authored patches may store other settings, and Chorus Hiss at 100% retains the datasheet maximum.

The 34 musical front-panel controls have stable Reason sequencer-automation lanes. Preset Level is an internal browsing trim; Unit Character, Aging, Chorus Hiss, and the rear engine policies remain intentionally outside sequencer automation.

Clicking the already-selected Poly 1, Poly 2, or Unison button repeats the instrument's held-key reassignment gesture.

Rear panel and engine quality



The rear panel provides monophonic Note and Gate CV inputs, stereo audio outputs, four song-persistent engine policies, and the patch-persistent Unit Character control plus the separate song-persistent Aging control. Patch browsing never changes the four engine-policy settings or Aging; patches may set Unit Character. The four engine policies remain available to Remote but are deliberately omitted from sequencer automation. Character and Aging are rear-only, available to Remote, and omitted from sequencer automation. The compact front strip has six read-only cells reporting the four engine policies plus Character and Aging; the editable Character/Aging controls remain on the rear. Because those four policies change CPU use or the filter kernel, assign them to hardware controls deliberately.

Unit Character varies calibrated circuit tolerances across a 0-200% range. It is a patch-level sound setting, so it can be saved and recalled with a patch without creating a sequencer-automation lane.

Aging is a distinct rear-only 0-100% control stored with the song rather than a patch. New devices start at 0%, the freshly serviced state. Raising it models the source instrument's documented service drift: per-voice filter trims move by their individual share of up to a quarter tone and the noise source rises by up to 3.5 dB. Aging is available to Remote but deliberately omitted from sequencer automation.

- **Quality - 1x, 2x, 4x.** New devices use 1x. The selection is a requested ceiling: higher settings increase the internal rate and CPU cost unless an already-high host rate meets the bandlimiting target. A change waits until voices and output tails are quiet, then crosses through a short safety fade. Use 4x for offline or lightly loaded work; 16 voices at 4x is intentionally the heaviest combination.

- **VCF Tanh - Exact, Fast, Poly.** Poly is the efficient default and enables silent-card retirement. Fast uses the previous lookup kernel. Exact restores the continuously rendered reference path and is CPU-heavy.
- **Fast Early - Hermite, Cubic.** Cubic is the efficient default used with Poly; Hermite retains the earlier interpolation path.
- **VCF Solver - Max, High, Normal.** Normal is the efficient default. High and Max spend progressively more arithmetic on each internal filter sample.

For a fair CPU comparison, create a new device or explicitly select 1x, Poly, Cubic, and Normal. Existing songs correctly retain their saved choices.

Patches

The bank contains 77 original Protocodus patches: Init, 18 featured sounds, and 58 categorized additions. The newest 30 are clean-room universal bass, brass, string, and pad sounds. A patch-only calibration trim keeps browsing levels balanced without moving the visible Volume control. Downward attenuation applies immediately when a patch changes; upward makeup ramps briefly to avoid clicks. Reason's metadata and tags make all patches searchable by sound type and character.

The Rack Extension does not expose MIDI Program Change or SysEx file import/export; use Reason's patch browser and patch files instead.

CV, tuning, and timing

The rear Note/Gate CV pair is monophonic; MIDI remains polyphonic up to the selected Voices limit. Gate CV triggers a note and Note CV sets its pitch. Changing Note CV while Gate remains high retargets the held note legato. The instrument follows Reason's master tuning preference and reports a fixed 41-sample latency at every Quality setting.

Reason's audio-reset request clears voices, envelopes, chorus state, deferred quality changes, and wrapper note bookkeeping. Transport stop releases sounding notes normally.

Troubleshooting

- **Notes arrive but there is no sound:** load Init or reset the device, verify Volume and VCA Level, and confirm that stereo outputs are connected. Current builds include the preset-switch held-key reassignment fix.
- **High CPU:** select 1x, Poly, Cubic, and Normal on the rear. Reduce the Voices setting for dense arrangements. Reserve 4x with many voices for offline or lightly loaded work.
- **A Quality change seems delayed:** release all notes and let the output tail settle; the change is deliberately applied only at a safe silent boundary.
- **No CV response:** connect both Gate and Note CV, then verify the source uses Reason's normal pitch and gate ranges.
- **A problem persists:** visit protocodus.cz/product/youknow and include the YouKnow version, Reason version, operating system, computer model, audio sample rate, patch name, and reproduction steps in a message to protocodus@proton.me.

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Privacy notice

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Audio and control data are processed inside Reason on the user's computer. Reason may store YouKnow settings in songs and patches. Reason Studios handles accounts, licensing, purchases, downloads, host diagnostics, and any related network activity under its own settings and privacy terms.

If a user contacts support, Protocodus receives the information they choose to send, such as an email address, diagnostic details, or attachments. It is used to answer the request, investigate the issue, meet legal obligations, and protect the service. Users should not send confidential songs, credentials, or recordings unless they are necessary and safe to share.

Privacy questions and requests can be sent to protocodus@proton.me. Product and company information is available at the YouKnow support page. An updated notice will accompany a release if YouKnow's data practices change.

Third-party notices and provenance

Protocodus' original YouKnow source and panel artwork are covered by the MIT license. The Rack Extension is built with the separately licensed Reason Rack Extension SDK and is intended for distribution through the authorized Reason Studios channel. Reason SDK material is not relicensed under YouKnow's MIT license.

The pitch and modulation wheels, audio and CV sockets, device-name tape, and back-panel placeholder use the SDK's standard instrument artwork so they behave and read like native Reason controls. The rear mono and spreading symbols are the white variants from Reason Studios' RE2D Stock Graphics 1.1 pack. These Reason assets remain covered by the SDK and stock-graphics terms, not YouKnow's MIT license. Panel text is rasterized during development from Apple-supplied DIN fonts; the font files themselves are not shipped in the Rack Extension.

Modelling references

YouKnow is an independent implementation informed by published virtual-analog research. It does not copy or include the reference implementations associated with these publications.

- Vadim Zavalishin, *The Art of VA Filter Design* - topology-preserving transforms and ladder-filter analysis.
- Tim Stilson and Julius O. Smith, *Analyzing the Moog VCF with Considerations for Digital Implementation* (1996) - ladder root-locus analysis.
- Antti Huovilainen, *Non-linear digital implementation of the Moog ladder filter* (DAFx-04), and Stefano D'Angelo and Vesa Valimaki, *Generalized Moog Ladder Filter: Part II* - nonlinear delay-free-loop methods.
- Vesa Valimaki, Jussi Pekonen and Juhan Nam, *Perceptually informed synthesis of bandlimited classical waveforms using integrated polynomial interpolation* (JASA, 2012) - bandlimited waveform synthesis.
- Martin Holters and Julian Parker, *A Combined Model for a Bucket Brigade Device and its Input and Output Filters* (DAFx-18) - bucket-brigade delay modelling.

Patches

The shipped bank contains 77 patch names and parameter states created for YouKnow by Protocodus. No third-party factory tone-memory records or archival factory labels are included.

No firmware, ROM image, sample, impulse response, captured audio, or separately sold sound-bank data is included.

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